ECE 310 Fall 2026
Lecture 17: Ideal Analog-to-Digital Conversion

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3583 2479

Corey Snyder

University of Illinois Urbana-Champaign

October 9th, 2026



Learning Objectives

After this lecture, you should be able to:

- State and sketch the resulting DTFT of a discrete-time signal obtained via ideal sampling of a continuous-time signal.
- Demonstrate how changes to the sampling rate will affect the resulting DTFT of the obtained discrete-time signal.
- Explain Nyquist criterion for sampling continuous-time signals.
- Identify the Nyquist rate of a continuous-time signal.



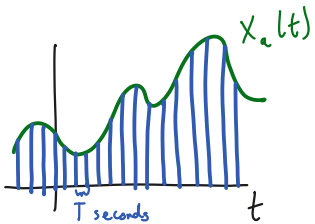
Ideal Sampling of Analog Signals

Recall from Lecture 1 that ideal sampling of continuous signals $x_a(t)$ is expressed in the time-domain as

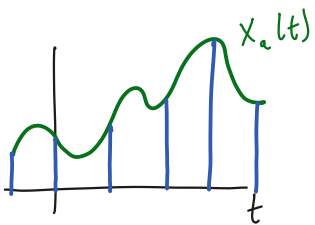
$$x[n] = x_a(nT), \ -\infty < n < \infty.$$

T = sampling period $\quad \frac{1}{T} = f_s$ = sampling frequency

How do we choose T?



T small

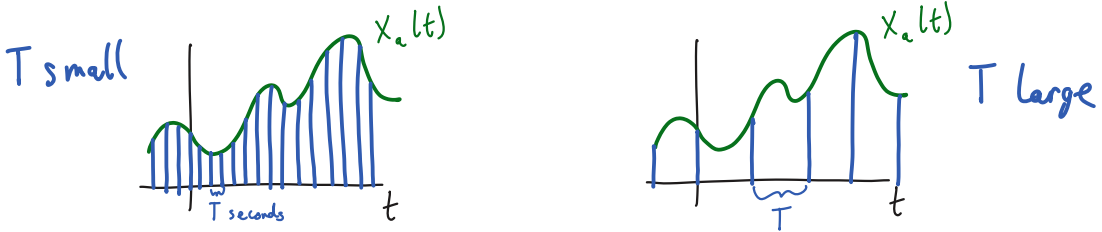


T Large

Ideal Sampling of Analog Signals

Recall from Lecture 1 that ideal sampling of continuous signals $x_a(t)$ is expressed in the time-domain as

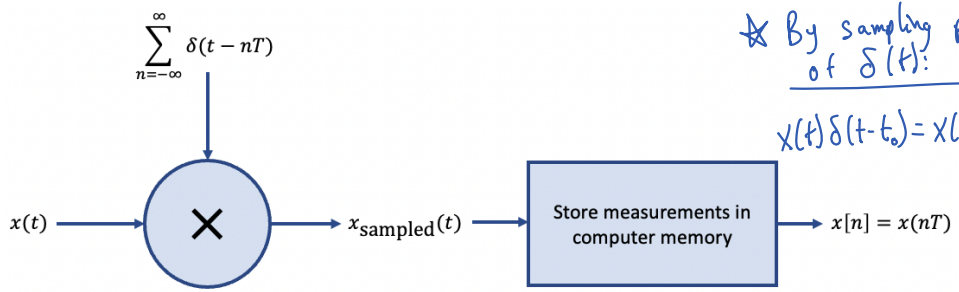
$$x[n] = x_a(nT), \ -\infty < n < \infty.$$



Key question of today's lecture

How does the choice of sampling period T affect the representation of $x[n]$? Are there certain safe choices of T that guarantees we can recover $x_a(t)$ from $x[n]$?

Ideal Sampling of Analog Signals



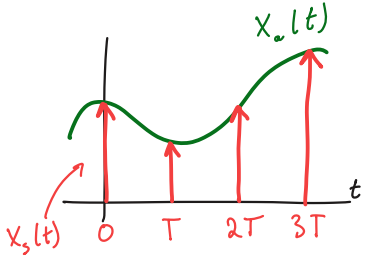
★ By sampling property of $\delta(t)$:

$$x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$$

$$x_s(t) = x_{\text{sampled}}(t) = x_a(t)\sum_{n=-\infty}^{\infty}\delta(t-nT)$$

$$\overset{\bigstar}{=} \sum_{n=-\infty}^{\infty} x_a(t)\,\delta(t-nT)$$

$$= \sum_{n=-\infty}^{\infty} x_a(nT)\,\delta(t-nT)$$



Ideal Sampling of Analog Signals

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$$x_s(t) = x_a(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = X_a(t)\, f(t)$$

$$X_s(\Omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T} n\right)$$

Windowing/Multiplication in time property

$$X_a(t)\, f(t) \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} X_a(\Omega) * F_a(\Omega)$$

$\rightarrow x_s(t) \xleftrightarrow{\mathcal{F}} X_s(\Omega)$

$\rightarrow x_a(t) \xleftrightarrow{\mathcal{F}} X_a(\Omega)$

$\rightarrow f(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \xleftrightarrow{\mathcal{F}} \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta\left(\Omega - \frac{2\pi}{T} n\right)$

$$X_s(\Omega) = \frac{1}{2\pi} X_a(\Omega) * F_a(\Omega)$$

$$= \frac{1}{2\pi} \left(X_a(\Omega) * \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta\left(\Omega - \frac{2\pi}{T} n\right) \right)$$

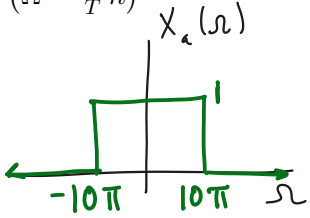
$$= \frac{1}{T} \sum_{n=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T} n\right)$$

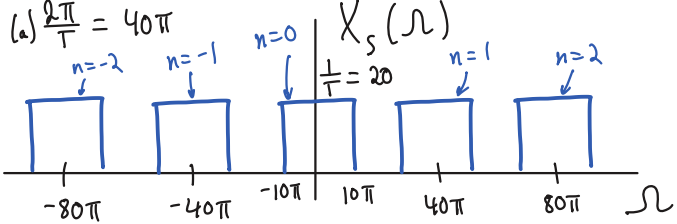
Example 1: Varying T

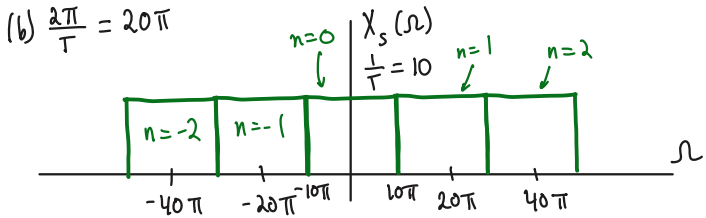
Let $X_a(\Omega) = \text{rect}\left(\frac{\Omega}{20\pi}\right)$. Remember $X_s(\Omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T}n\right)$

(a) Sketch $X_s(\Omega)$ for $T = \frac{1}{20}$ s for $-100\pi < \Omega < 100\pi$.

(b) Sketch $X_s(\Omega)$ for $T = \frac{1}{10}$ s for $-50\pi < \Omega < 50\pi$.







Ideal Sampling of Analog Signals

$$X_s(\Omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T} n\right)$$

$$X_d(\omega) = X_s(\Omega)\Big|_{\Omega = \frac{\omega}{T}} = \frac{1}{T} \sum_{n=-\infty}^{\infty} X_a\left(\frac{\omega - 2\pi n}{T}\right)$$



→ How are Ω and ω related?

analog ↗ ↖ digital

$\omega = \Omega T$

→ Let $x_a(t) = \cos(\Omega_0 t)$

$$x[n] = x_a(nT) = \cos(\Omega_0 nT)$$
$$= \cos(\Omega_0 T n)$$
$$= \cos(\omega_0 n)$$

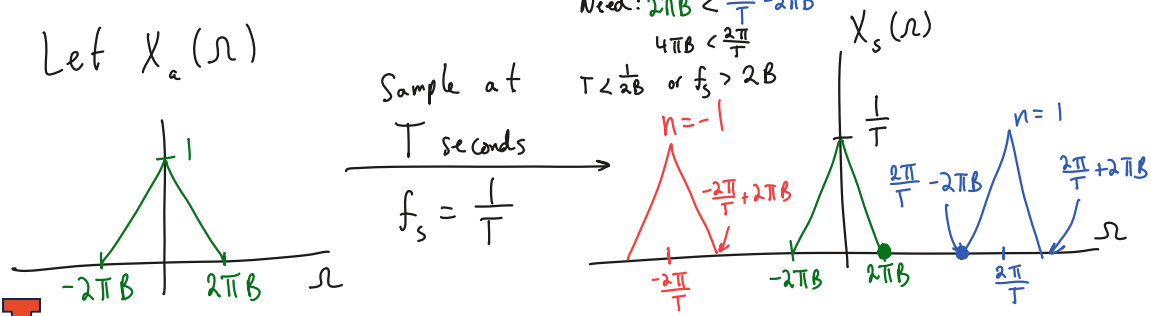


What choice of T works?

Bandlimited signals

A *bandlimited* continuous-time signal $x_a(t) \overset{\mathcal{F}}{\longleftrightarrow} X_a(\Omega)$ has bandlimit B (in Hz) or $2\pi B$ (in radians/second) such that

$$|X_a(\Omega)| = 0, \ |\Omega| \geq 2\pi B.$$



Nyquist Criterion

Nyquist criterion

A **bandlimited** continuous-time signal with maximum linear frequency B can be sampled at sampling period T to produce $x[n]$ such that $x_a(t)$ may be recovered from $x[n]$ if

$$T < \frac{1}{2B}$$
$$f_s > 2B$$

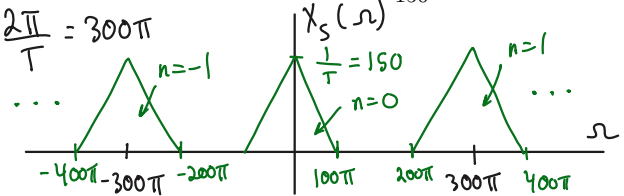
Otherwise, we will incur *aliasing* when obtaining $x[n]$. The frequency $f_{\text{nyquist}} = 2B$ (units in Hz) is known as the *Nyquist rate*.



Example 2: Sampling Around f_{nyquist}

Let $X_a(\Omega) = \begin{cases} 1 - \frac{|\Omega|}{100\pi}, & |\Omega| \le 100\pi \\ 0, & |\Omega| > 100\pi \end{cases}$. Recall $X_d(\omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a\left(\frac{\omega - 2\pi n}{T}\right)$ and $\omega = \Omega T$.

(a) Sketch $X_d(\omega)$ when $T = \frac{1}{150}$ s.



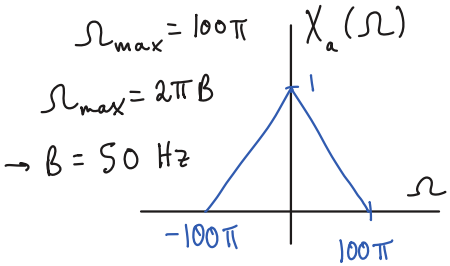
$\rightarrow f_{\text{nyquist}} = 2B = 100$ Hz

$\rightarrow T_{\text{nyquist}} = \frac{1}{100}$ s

$\rightarrow$ Need $T < T_{\text{nyquist}}$

$\rightarrow X_s(\Omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T}n\right)$

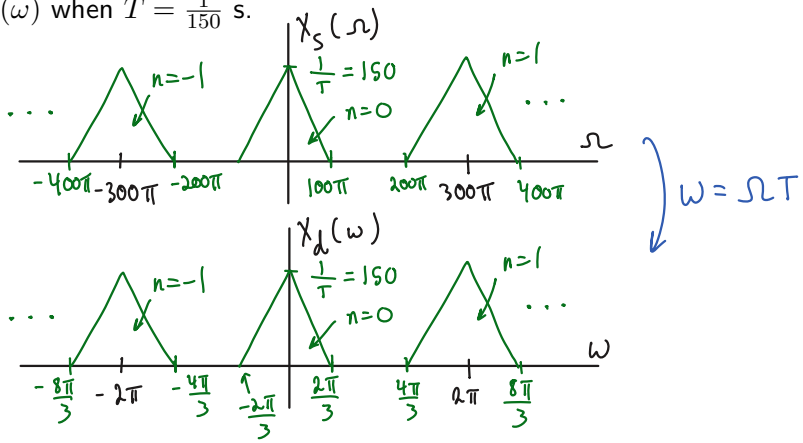
$\rightarrow \omega = \Omega T$



Example 2: Sampling Around f_{nyquist}

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(a) Sketch $X_d(\omega)$ when $T = \frac{1}{150}$ s.

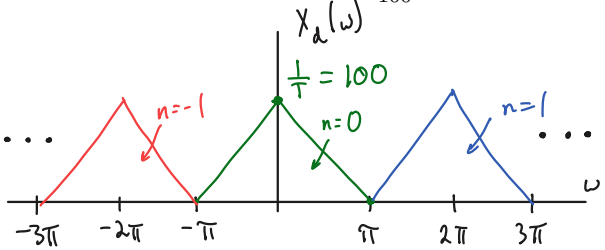


Example 2: Sampling Around f_{nyquist}

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→ $X_s(\Omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T}n\right)$

(b) Sketch $X_d(\omega)$ when $T = \frac{1}{100}$ s.

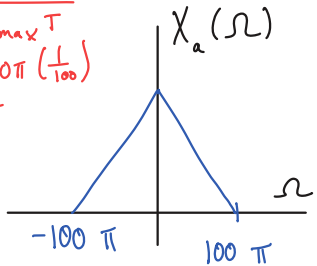


For central copy: → $\omega = \Omega T$

$\omega_{max} = \Omega_{max} T$

$= 100\pi\left(\frac{1}{100}\right)$

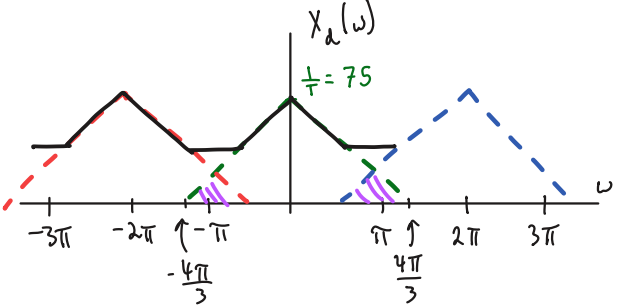
$= \pi$



Example 2: Sampling Around f_{nyquist}

Let $X_a(\Omega) = \begin{cases} 1 - \frac{|\Omega|}{100\pi}, & |\Omega| \leq 100\pi \\ 0, & |\Omega| > 100\pi \end{cases}$. Recall $X_d(\omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a\left(\frac{\omega - 2\pi n}{T}\right)$ and $\omega = \Omega T$.

(c) Sketch $X_d(\omega)$ when $T = \frac{1}{75}$ s.



$\rightarrow X_s(\Omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T}n\right)$

$\rightarrow \omega = \Omega T$

$\rightarrow \omega_{max} = \Omega_{max} T$
$= 100\pi \cdot \frac{1}{75}$
$= \frac{4\pi}{3}$

